

Topic 3: Bonding and Structure

3A: Ionic bonding

Students will be assessed on their ability to:

3.1	know and be able to interpret evidence for the existence of ions, limited to physical properties of ionic compounds, electron density maps and the migration of ions
3.2	be able to describe the formation of ions in terms of loss or gain of electrons
3.3	be able to draw dot-and-cross diagrams to show electrons in cations and anions
3.4	be able to describe ionic crystals as giant lattices of ions
3.5	know that ionic bonding is the result of strong net electrostatic attraction between ions
3.6	understand the effects of ionic radius and ionic charge on the strength of ionic bonding
3.7	understand reasons for the trends in ionic radii down a group in the Periodic Table, and for a set of isoelectronic ions, including N^{3-} to Al^{3+}
3.8	understand the meaning of the term 'polarisation' as applied to ions
3.9	understand that the polarising power of a cation depends on its radius and charge, and the polarisability of an anion also depends on its radius and charge
	Further suggested practical The migration of ions in a U-tube using copper(II) chromate solution or on a microscope slide using potassium manganate(VII) crystals

3B: Covalent bonding

Students will be assessed on their ability to:

3.10	understand that covalent bonding is the strong electrostatic attraction between two nuclei and the shared pair of electrons between them, based on the evidence: i the physical properties of giant atomic structures ii electron density maps for simple molecules
3.11	be able to draw dot-and-cross diagrams to show electrons in covalent substances, including: i molecules with single, double and triple bonds ii species with dative covalent (coordinate) bonds, including Al_2Cl_6 and the ammonium ion
3.12	be able to describe the different structures formed by giant lattices of carbon atoms, including graphite, diamond and graphene, and discuss the applications of each
3.13	understand the meaning of the term 'electronegativity' as applied to atoms in a covalent bond

3.14	know that ionic and covalent bonding are the extremes of a continuum of bonding type and be able to explain this in terms of electronegativity differences, leading to bond polarity in bonds and molecules, and to ionic bonding if the electronegativity is large enough
3.15	be able to distinguish between polar bonds and polar molecules and predict whether or not a given molecule is likely to be polar
	Further suggested practical Determine the effect of an electrostatic force on jets of liquids (water, ethanol and cyclohexane) and use the results to determine whether the molecules are polar or non-polar

3C: Shapes of molecules

Students will be assessed on their ability to:

3.16	understand the principles of the electron-pair repulsion theory, used to interpret and predict the shapes of simple molecules and ions
3.17	understand the terms 'bond length' and 'bond angle'
3.18	know and be able to explain the shapes of, and bond angles in, BeCl_2 , BCl_3 , CH_4 , NH_3 , NH_4^+ , H_2O , CO_2 , gaseous PCl_5 , SF_6 and C_2H_4
3.19	be able to apply the electron-pair repulsion theory to predict the shapes of, and bond angles in, molecules and ions analogous to those in 3.18

3D: Metallic bonding

Students will be assessed on their ability to:

3.20	understand that metals consist of giant lattices of metal ions in a sea of delocalised electrons
3.21	know that metallic bonding is the strong electrostatic attraction between metal ions and the delocalised electrons
3.22	be able to use the models in 3.20 and 3.21 to interpret simple properties of metals, including electrical conductivity and high melting temperature